

Day – 7 VLAN Concept Theory.

Task – 1 Name the VLANs using commands.

- Step-by-Step Configuration:- Click on the Switch in your workspace, go to the CLI tab, and press Enter to get the prompt.
- Enter privileged EXEC mode and open global configuration mode:
- enable
- configure terminal
- Create and name VLAN 10:
- vlan 10
- name Engineering
- exit
- Create and name VLAN 20:
- vlan 20
- name Marketing
- exit
- Exit configuration mode and save your work:
- end

Task – 2 Assign the ports to VLANs

- Step 1: Assign Ports to VLAN 10 (Engineering):- Open your switch CLI, enter global configuration mode, and run the following commands:
- configure terminal
- interface range FastEthernet0/1 - 2
- switchport mode access
- switchport access vlan 10
- Exit
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- Step 2: Assign Ports to VLAN 20 (Marketing):- Next, configure ports FastEthernet0/3 and FastEthernet0/4 for VLAN 20:
- interface range FastEthernet0/3 - 4
- switchport mode access
- switchport access vlan 20
- exit

- end
- Step 3: Verify the Port Assignments:- To confirm that the ports are correctly mapped to their VLANs, run this verification command in privileged EXEC mode:
- show vlan brief

Task – 3 Config the VLANs IDs.

- VLAN Configuration :- This document outlines the VLAN configuration implemented on the switch, detailing the VLAN IDs, names, and their respective assigned port memberships.
- VLAN 10: Engineering
- VLAN ID: 10
- VLAN Name: Engineering
- Assigned Ports: FastEthernet0/1, FastEthernet0/2
- Description: Dedicated network segment for the Engineering department, isolating their traffic from other departments.
- VLAN 20: Marketing
- VLAN ID: 20
- VLAN Name: Marketing
- Assigned Ports: FastEthernet0/3, FastEthernet0/4
- Description: Dedicated network segment for the Marketing department, ensuring broadcast containment and security between distinct business units.
- Default VLANs (Reference)
- VLAN 1: default (Active on remaining unassigned switch ports: Fa0/5 through gigabit uplink ports)

Task – 4 Benefits to work on VLANs

- 1. Enhanced Security and Isolation
- Preventing Accidental Disruption: Networking students frequently experiment with device configurations, routing protocols, and dynamic addressing tools (like setting up test DHCP servers). By separating the labs into different VLANs, any misconfiguration or rogue broadcast traffic generated during a networking practical will remain contained within its own logical segment.
- Protecting Student Work: The coding lab's environment stays completely isolated, preventing networking students from inadvertently sniffing packets, viewing traffic, or interfering with code repositories and test servers used by the coding students.

- 2. Broadcast Domain Control and Performance
- Reducing Unnecessary Traffic: Every time a device sends a broadcast message (like an ARP request or network discovery packet), every machine on that local network has to process it. Splitting the labs into separate VLANs creates distinct broadcast domains.
- Optimized Bandwidth: Heavy traffic or debugging loops in the networking lab won't flood the switch ports of the coding lab, ensuring that both groups enjoy stable, reliable network performance during classes.
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